

CATIA V5 ENVIRONMENT

SCOPE:

This manual describes CATIA version 5 working environment and the basic functions thereof.

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1 FOREWORD

The scope of this manual is to know and be familiarised with CATIA version 5 environment (from now on V5), as well as a description of the use of basic functions thereof.

2 MANAGING DOCUMENTS

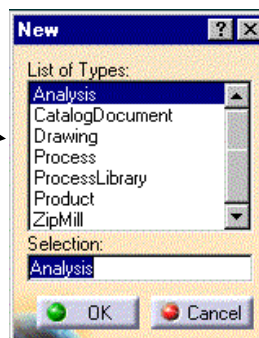
V5 is a CAD-CAM system operating under Windows environment. It owns the typical possibilities for managing documents i.e. open, save, delete, etc used in this type of environment.

2.1 Types of documents

V5 has several types of documents depending on their use:

- **Catpart:** Document containing information about 3D definition of the part.
- **Catproduct:** Document containing information about components that form an assembly as well as position thereof.
- **Analysis:** This document contains information related to every type of analysis.
- **Drawing:** Document containing information about 2D definition or drawing of a part or assembly.

List of types of documents



2.2 Create a document

There are several ways for creating a new document in CATIA V5. These are as follows:

- Using New option and File function.
- Using copy and paste utility of a document previously created.
- Choosing a Workbench in Start list.

2.3 Open a document

Whenever a document has to be opened, V5 provides several ways for it:

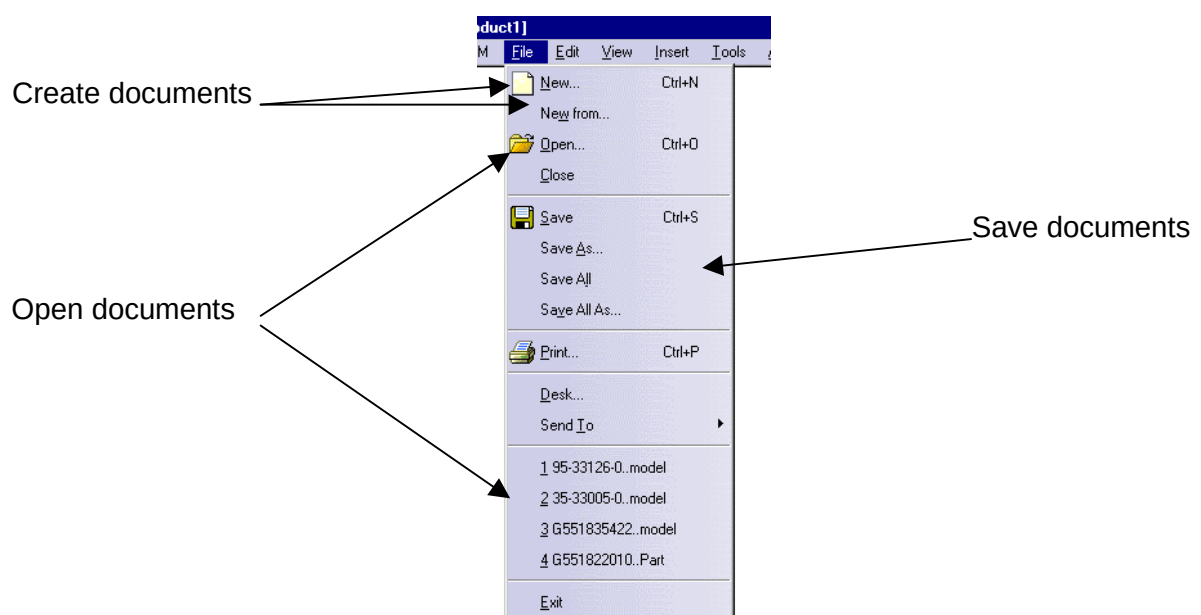
- Using **Open** option of **File** function
- Using a list of the last open documents existing in function **File**
- **Double-clicking** CATIA type files in Windows Explorer (NT only)

2.4 Save a document

Several options can be used of File function:

- **Save:** save an open document
- **Save As:** Save an open document with a different name
- **Save All:** save all open documents
- **Save All As:** save all open documents with different names

The following image shows File function with options



3 DOCUMENT DISPLAY

This paragraph includes the different types of display of a document in V5.

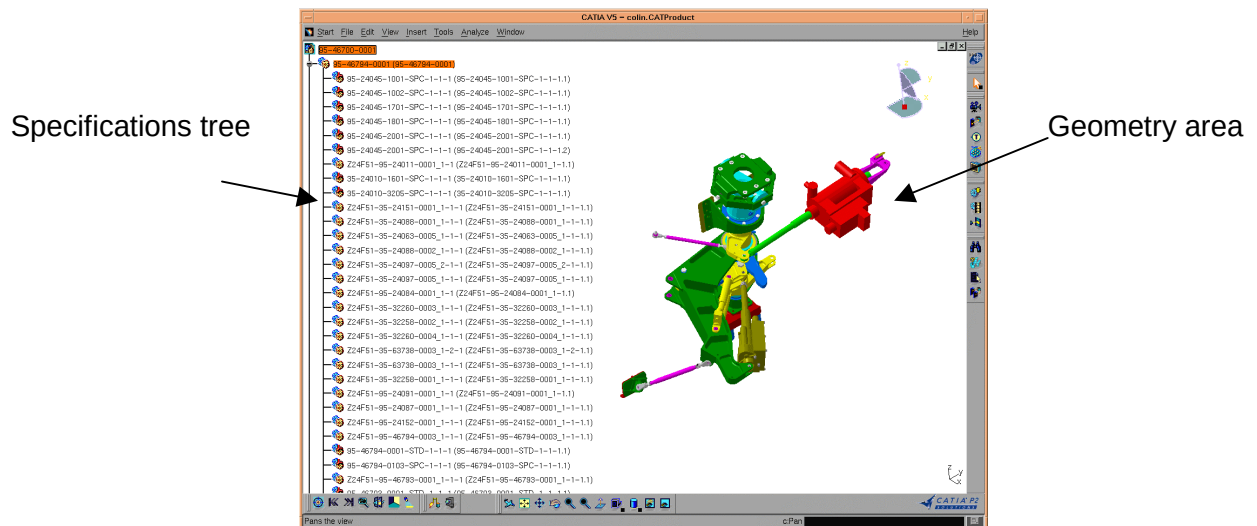
- Display areas
- Full screen
- Document in different windows
- Overview use

3.1 Display areas

Display of a document is carried out in two main areas: first area composed by the specification tree and second area showing geometry. Both areas are put together and it is possible at any time to work independently in any of them.

- **Specification tree:** In this area, specifications are shown in a tree shape with as many branches as different specifications the document has.
- **Geometry area:** area of creation and display of geometric components in V5.

The following image identifies both areas.



Hit **F3** function key to hide specifications tree. To change working area click onto tree for operating with tree or click axes appearing on the right low corner for operating with geometry.

3.2 Full screen

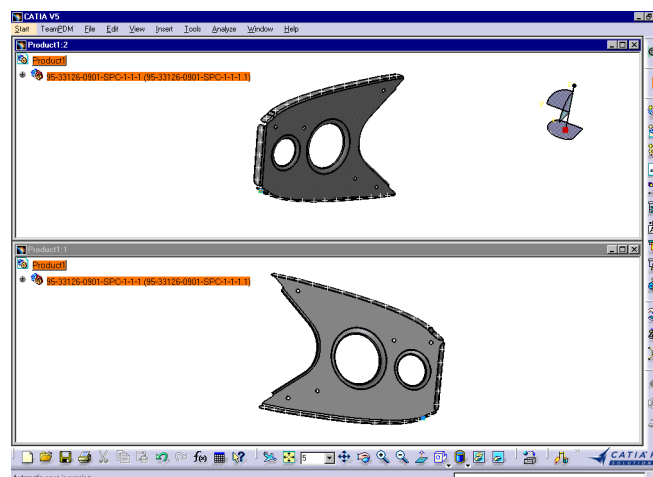
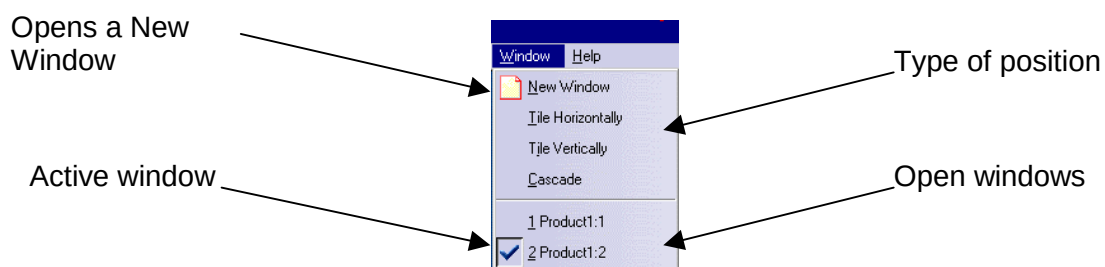
Whenever a document has to be displayed, it is possible to see the document at full screen, that is, specifications tree as well as geometry area will occupy the whole screen of the monitor, working area peripheral menus.

To arrange a full screen disposition select **Full Screen** option in **View** menu. Returning to the previous view is achieved right-clicking at any place of the screen and disabling Full Screen option.

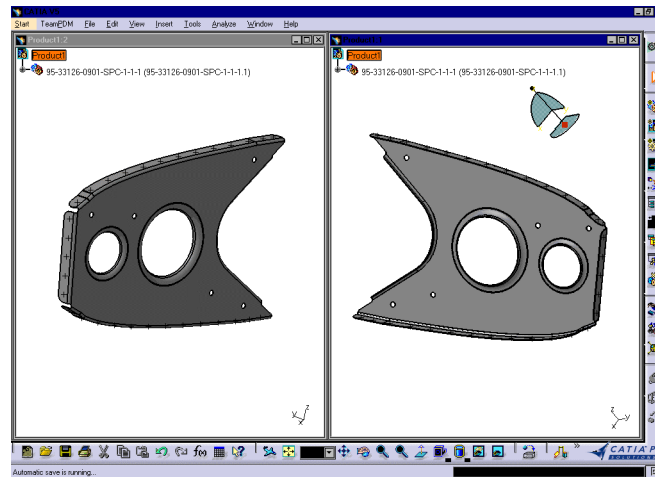
3.3 Document in different windows

A document open in a window can be displayed in a different window or windows at the same time, though just a single window can be active. Select **New Window** option from **Window** menu in order to open other windows.

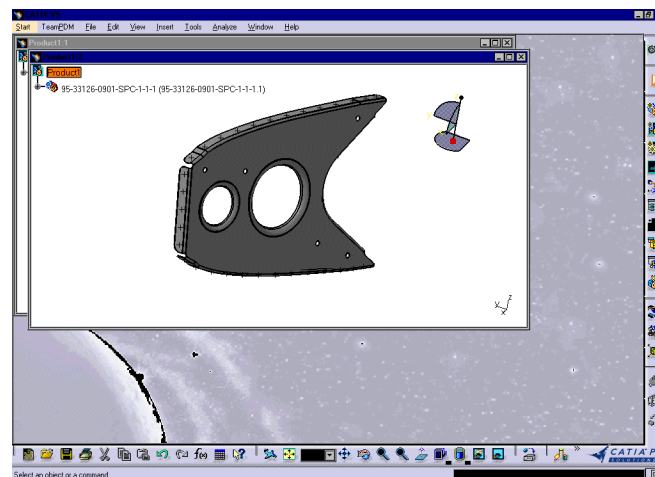
These windows can be positioned horizontally, vertically or in Cascade, using **Tile Horizontally**, **Tile Vertically** or **Cascade** respectively depending on the case, from **Window** menu.



Tile Horizontally windows



Tile vertically windows



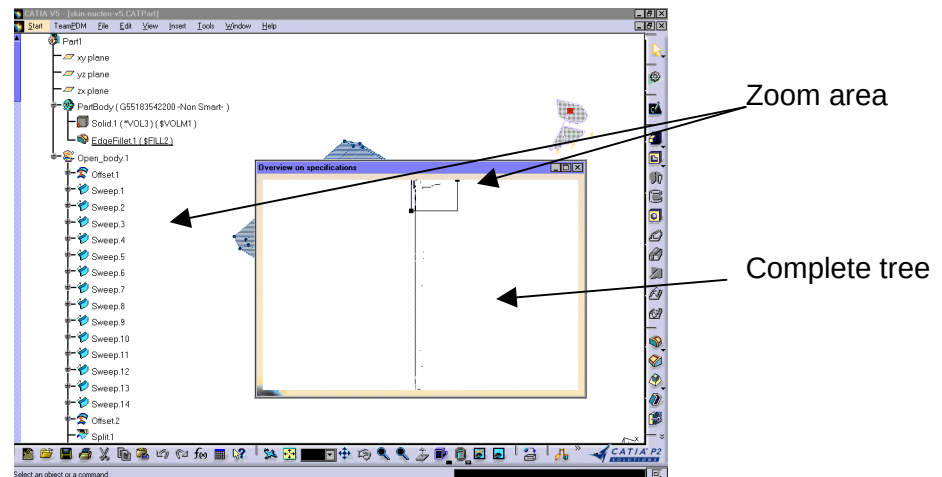
Cascade windows

Window menu as well as **Compass** indicates identification of the active window. An active window should show the **Compass**. To activate any other different window, just click onto it.

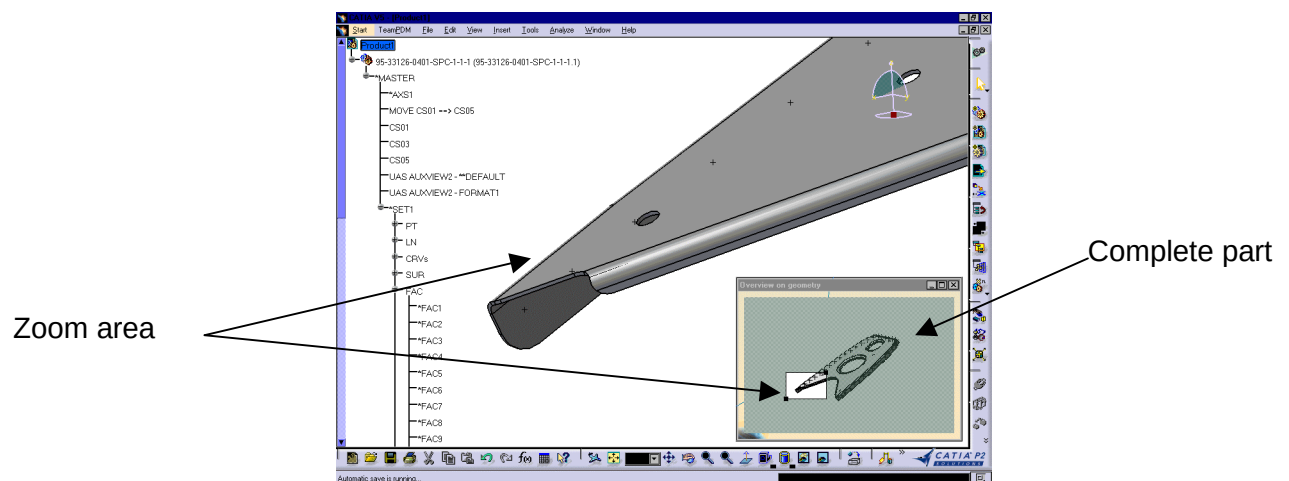
3.4 Use of Overview

Overview is used in cases in which both geometry and tree of specification are too big to observe some details. Overview is a zoom covering tree and geometry area,

- **Overview of specifications tree:** tree shall be visible to open its overview. Select **Specifications Overview** in View menu.



- **Overview of geometry area:** Select **Geometry** Overview option in View menu to open Overview of geometry area.

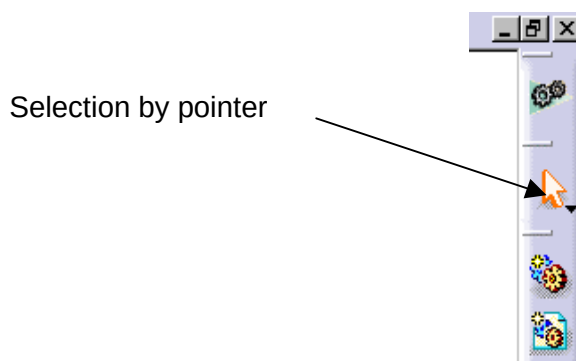


4 SELECTION OF OBJECTS

Selection of objects in V5 is performed using different tools as bounding outline, trap, magnifier or the use of different types of Search.

4.1 Use of pointer

Pointer selection has two main possibilities: pre-selection and selection. Both of them require selection mode as indicated in the following image:



- **Preselection:** It preselects and highlights directly and by default both fields, passing pointer on objects at the geometry area as well as at the specifications tree.
- **Selection:** Selection with pointer is performed clicking onto the element or object previously pre-selected.

Selection can be performed onto an object in the tree or geometry area. It is possible to select complete object or elements thereof, i.e. in a solid it is possible to select edges, faces, vertexes, etc.

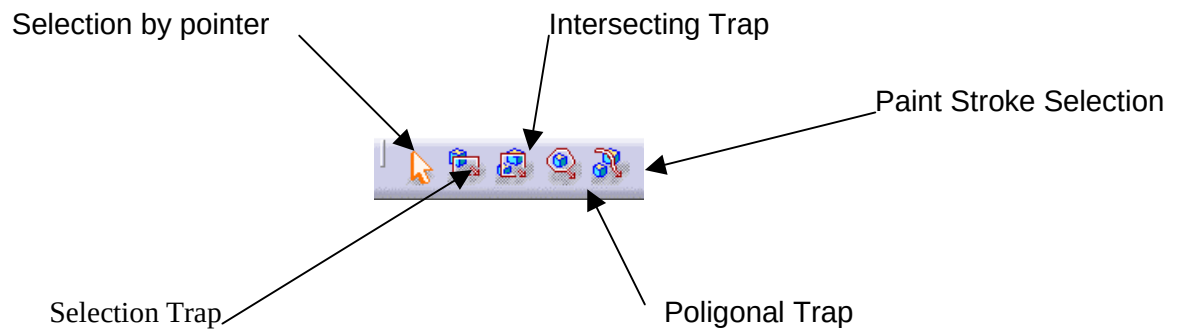
- **Multi-selection:** Using the pointer it can be performed holding down Control key and selecting objects or elements at the geometry area as well as at the specifications tree.

In the tree, besides the previous multiselection, it is possible to multi-select objects holding down Caps key.

- **De-selection:** De-selection of elements is performed if selected again.

4.2 Use of Trap

It is an outline selection type. It selects element inside outline or interfering it. Different types of Trap are:



- **Selection Trap:** Selects elements contained inside rectangle defined by points. It is generated by clicking in a point of the screen and dragging (holding down) to the other point. Selection mode can be in selection by pointer or Selection trap.
- **Intersecting Trap:** It selects only elements intersecting the rectangle. Rectangle creation process is the same as in Selection Trap, though it must only be in Intersecting Trap selection.
- **Polygonal Trap:** Using this selection mode, it is possible to select elements contained inside a polygon defined with points that are indicated on the screen. Double-clicking completes points selection.
- **Paint Stroke Selection:** Selects elements interfering with curve that has been defined clicking on the screen and dragging pointer through the course.

4.3 Use of magnifier

In cases in which due to proximity of elements or elements at the same place it is possible to use the magnifier option in order to make an easy selection of the suitable element.

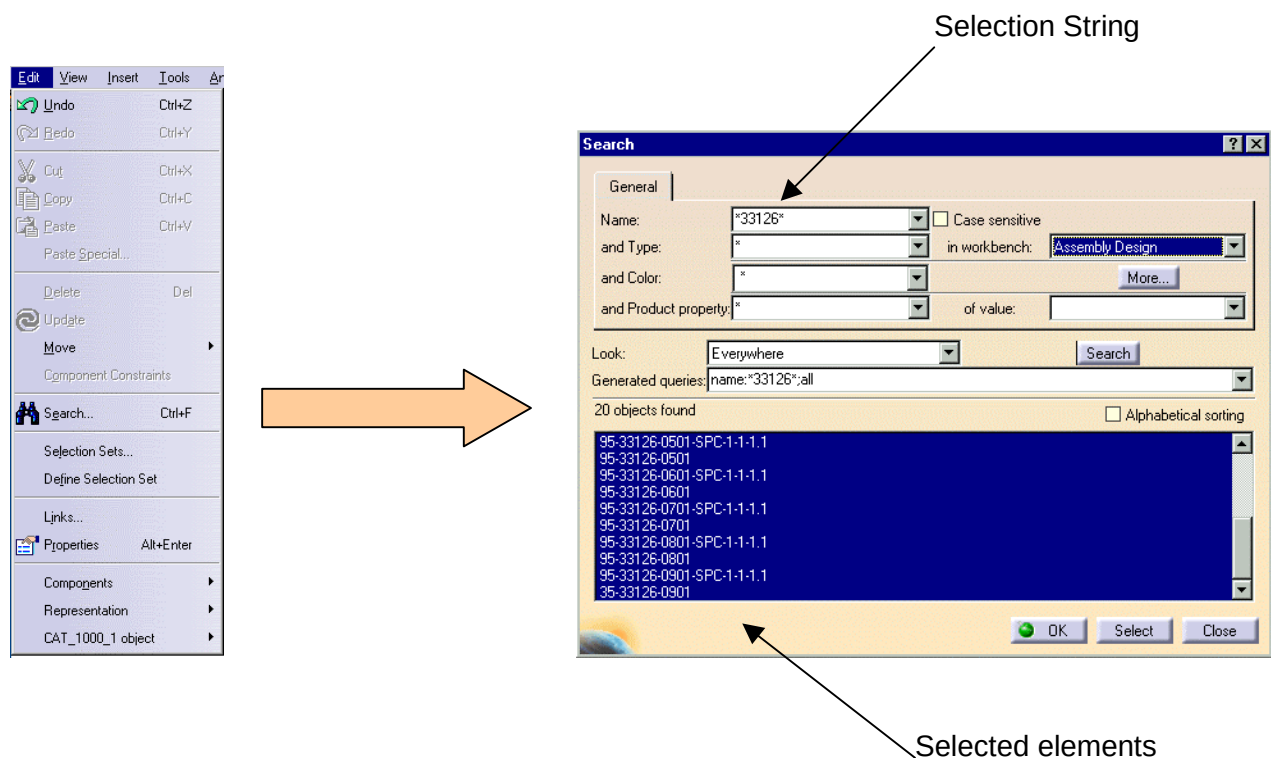
Magnifier discriminates between closed or overlaid elements allowing to choose the suitable element.

Magnifier is displayed when pointer is more than two seconds motionless onto an element, if indicated by visualization parameters. These parameters can be modified or the magnifier display removed in **Tools+Options+Visualization** in the paragraph **Display the preselection navigator after.....seconds**

4.4 Using Search option

Search selection mode selects by name filtering, type, color, etc, as well as a selection by the use of two or more previous filtering. For example, a CATPart document in which it is necessary to select all the elements of point type and red color.

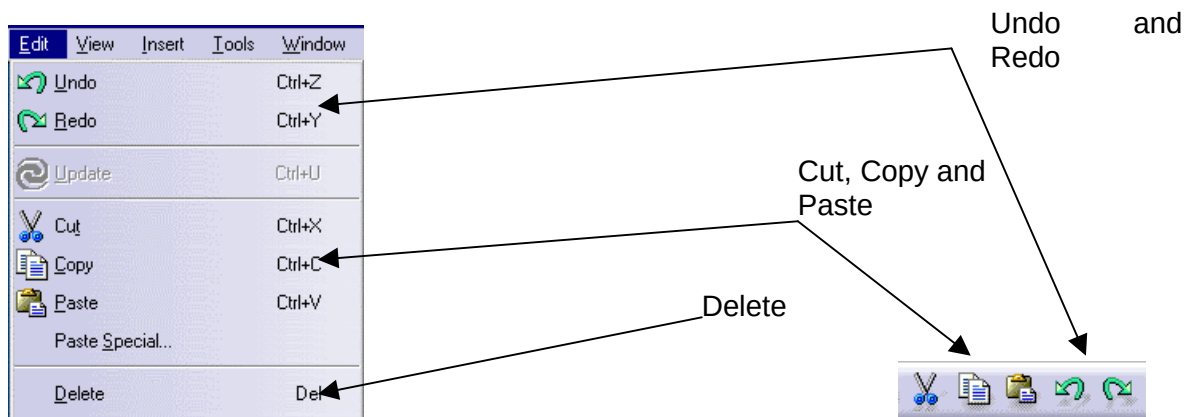
Search window is open selecting **Edit+Search**. The following images show an example of name filtering selection:



5 MANIPULATING OBJECTS

This chapter is about possibilities of manipulating objects in V5, typical possibilities of manipulating elements in Windows environment, as Undo, Redo, Cut, Paste, Properties, etc.

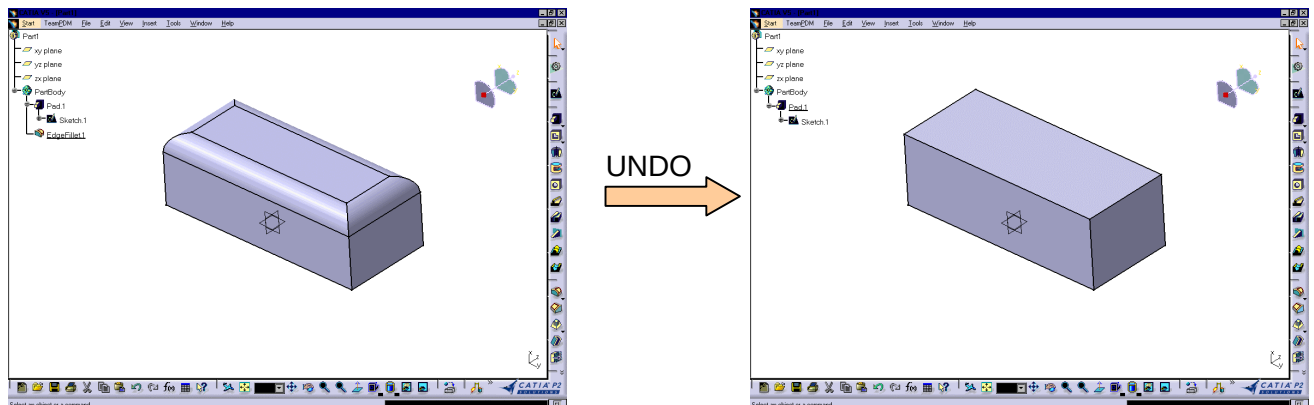
The following images show object manipulating icons for toolbar as well as Edit menu.



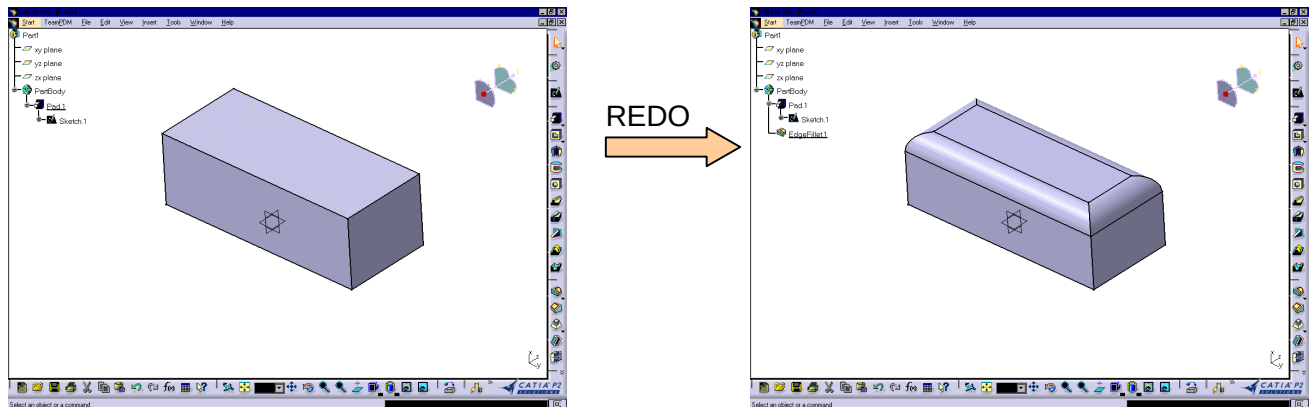
5.1 Use of Undo and Redo

Using Undo, it is possible to remove the last actions performed. Using Redo we removed the last Undo, returning to the state previous to Undo.

The following images show how by the use of Undo the last operation was removed from the solid, in this case the Fillet.



The following images show how by the use of Redo, the last Undo is removed and the last removed operation is recovered.



5.2 Cut, copy and paste objects

The use of Cut, Copy and Paste is the same as in Windows environment.

- **Cut:** Selecting an object and hitting this option, the selected object is deleted and maintained in the clipboard in order to be used with paste option.
- **Copy:** Equal to cut option but the selected object is not deleted.
- **Paste:** This option pastes objects included into the clipboard

5.3 Delete objects

To delete objects, it is necessary to select the object and hit Supr key, or select Delete option from Edit menu.

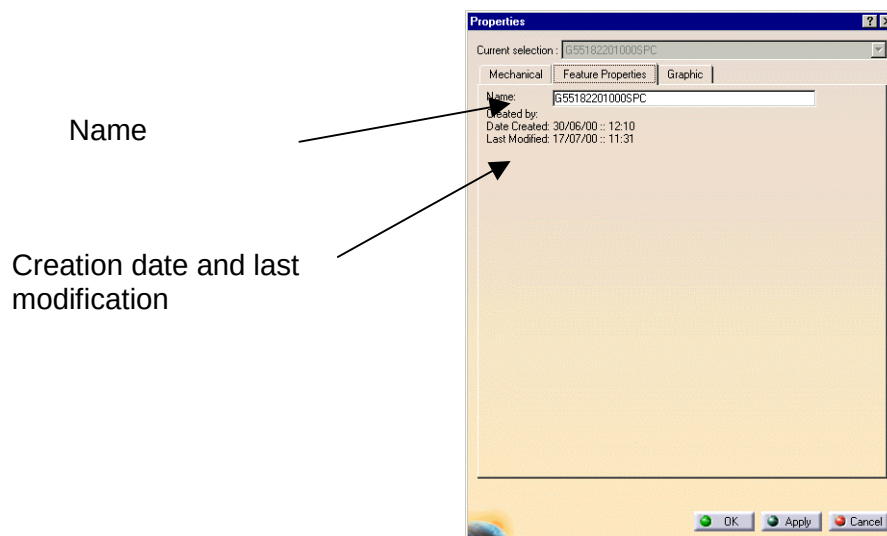
In case the object is connected with other objects, a window shall be displayed indicating relation and possibility of deleting related objects together with the selected object.

5.4 Properties of objects

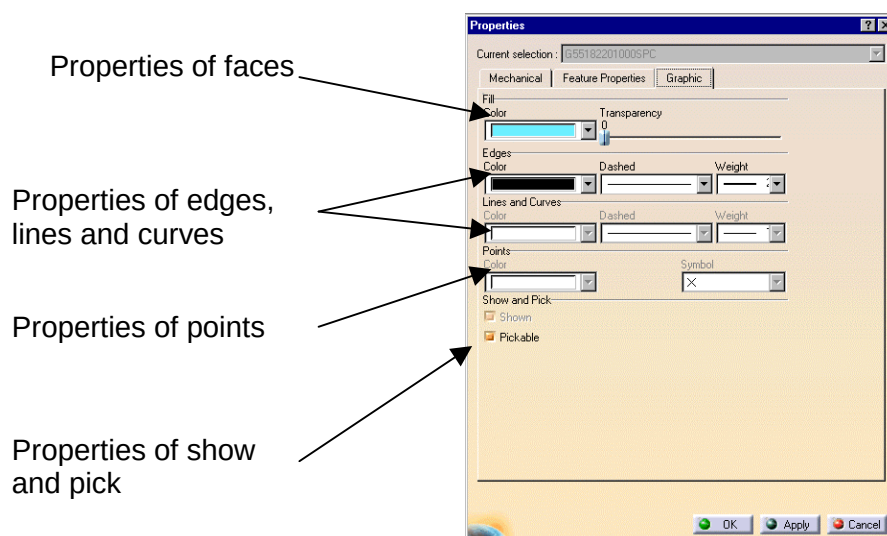
To display and modify properties of an object, it is necessary to select the object and choose Properties in Edit menu. It is also possible to open Properties window by right-clicking the mouse.

Properties window shall have different contents depending on the selected object. Nevertheless, Features Properties and Graphic paragraphs are common to all the objects.

- **Features Properties:** Information included in this paragraph – Properties window – is name of the object (it can be renamed), user who created it, creation date and last modification.



- **Graphics:** This paragraph of the Properties window displays and edits graphic properties of the object as Color and Transparency of faces, Color, Dashed and Weight of edges, lines and curves, Color and Symbol in points and Show and Pick attributes modification.



6 VISUALIZATION OF OBJECTS

This chapter shows the different means and tools of V5 in order to visualize objects.

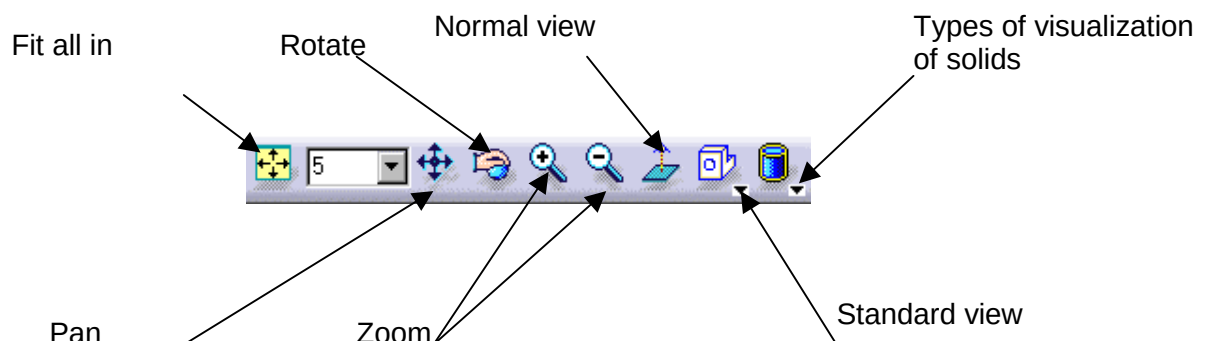
6.1 Use of the mouse in visualization

The mouse is the most important way to manipulate an image in V5. It is possible to move, rotate and zoom the image by the use of different buttons or combinations.

- **Panning:** By holding down mouse middle button and moving the mouse, a movement of translation is achieved on the screen.
- **Rotate:** By pressing and holding down middle button, press left button and moving mouse with buttons pressed, a rotation movement is achieved.
- **Zoom:** By pressing and holding down middle button, click left button and moving up and down the mouse, a zoom of the image is achieved.

6.2 Visualization Tools

V5 visualization tools are the following:

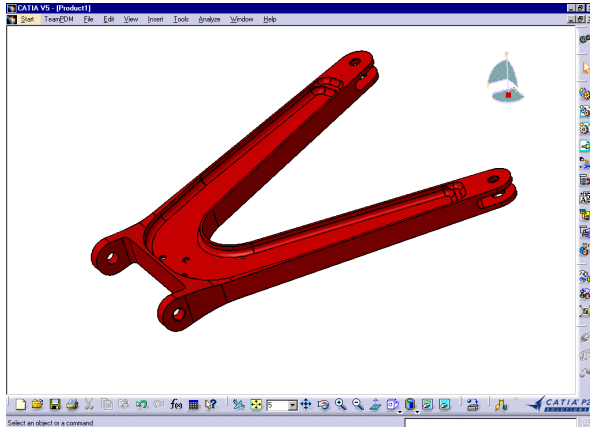


6.3 Views

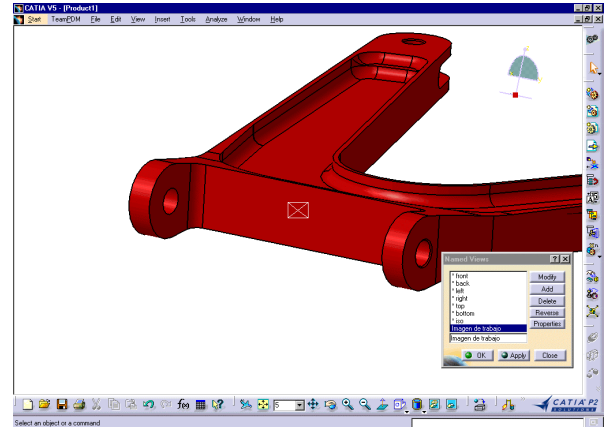
This paragraph shows how it is possible to generate and visualize non-standard views i.e. views that are created by the user.

Creation: Creation of views in V5 is performed using Named Views in View menu. The process is as follows:

- Locate desired view using rotation, zoom or panning.
- Open Named Views window and select Add.
- Change name of the view.



Initial view



Defined view

Visualization: To visualize views created by the user, it is necessary to open Named Views window and double-click onto the view to visualize.

Standard views

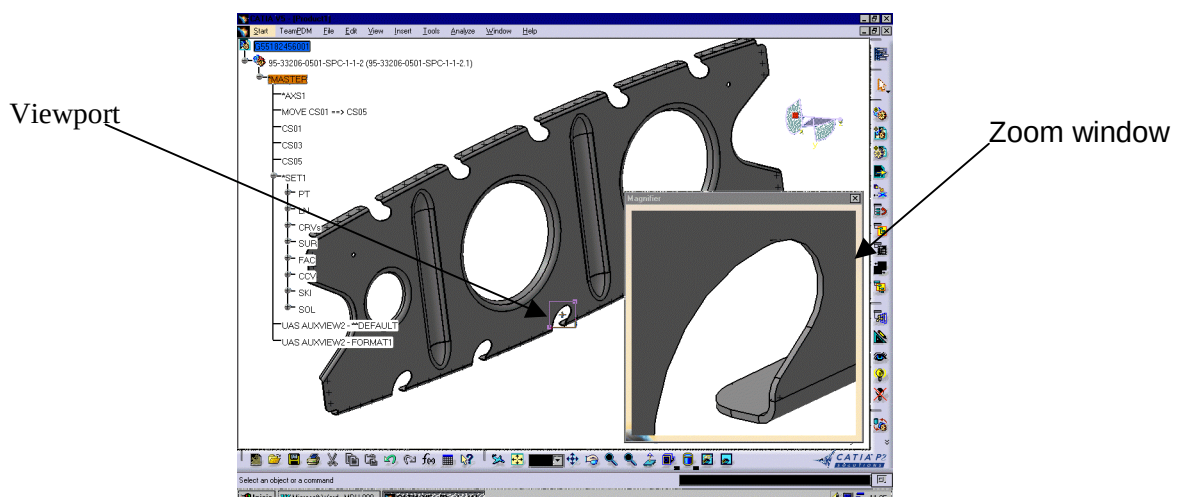
Defined views



6.4 Zoom window

Using zoom window, a zoom of a geometry area – previously marked with a viewport - is carried out. This zoom is shown in a new window.

To generate zoom window, select Magnifier option in View menu.



Creating a zoom window, viewport shall appear –by default- at the centre of the screen. Selecting and dragging centre point with cursor, it will be possible to move within the screen, moving the image within zoom window.

Likewise, if any handle is selected and dragged it will be possible to increase or decrease the viewport, obtaining an magnification or reduction of the image in the zoom window.

7 LAYERS

The existing V5 layers structure goes from 0 layer up to 999 layer. From 1 to 999 are to be previously enabled.

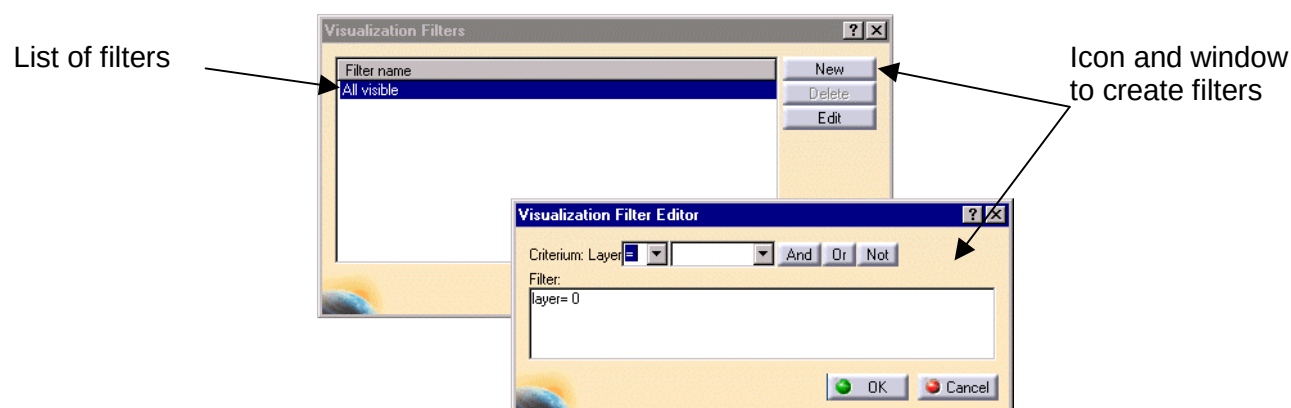
None option allows elements to be out of layers to avoid possible filtering (always visible).

- **None Layer:** Without layer, elements of this option have no associated layer.
- **General 0 Layer:** Layer created by default.
- **Layers from 1 to 999:** Possible layers to be activated or enabled.

Layer where an element is generated will be assigned to it. To change a layer of an element, it has to be selected and a new layer chosen.

7.1 Layer Filters

Creation of filters as well as application thereof is performed using Visualization filters option in Options menu. Selecting this option a new window will be open:



8 SHOW AND NO SHOW

This chapter explains how to hide or show objects and visualize No show space

Hide : Steps for hiding objects are the following:

- Select object to hide
- Select Hide/Show icon

Show: Steps are the same as for hiding but performing operation from No Show space.

No show space visualization: To visualize No show space, select icon transfer icon from Show space to No show space.

Transfer objects
from show to
hide and from
hide to show



Transfer between
Show space and
No show

References

Glossary

MDU

User's manual

Approval and authorisation

	Approval	Authorisation

Record of revisions

Issue	Date	Summary and cause of modification